Northstar Exception Response Center

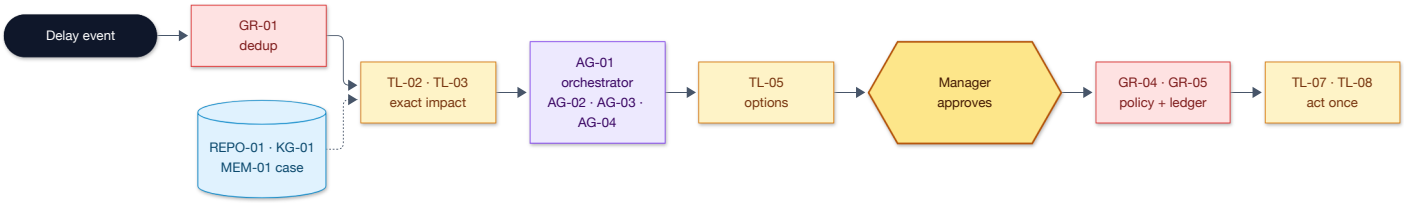
Owner: Chief Operating Officer · **Version:** 1.0 · **Date:** 2026-11-04-17-00-00 UTC · Example 4 of 4, capstone (illustrative numbers)

1. Problem

> **Operations managers** struggle to **respond to supply exceptions before customer orders are hit** because **logistics, inventory, purchasing and support each see one part of the impact and coordinate by email**, resulting in **≈ 26 h from delay to plan and 18 % of affected priority orders missing their SLA (≈ €120k a year in penalties)**.

Scope: inbound delays and stock breaches, from detection to an approved internal action. **Non-goals:** external purchases, payments, auto-sent customer messages.

2. Solution: events trigger, code computes, agents propose, a manager approves, the workflow acts once



Component	Used for	Catalog ID
Records + impact graph + durable case state	Canonical quantities; who is affected; resume after failure	REPO-01, KG-01, MEM-01
Tools	Read, compute options, propose, act (approved, idempotent), draft notices	TL-01…09
Models	Orchestration and drafts (GenAI); revised-ETA forecast (task-specific)	MOD-01, MOD-02
Workflow / agents	Durable detect → act → recover; orchestrator + 3 read-only specialists	WF-01, WF-02, AG-01…04

#	Use case	Target user	Priority
UC1	Delayed-shipment impact analysis	Planner	High
UC2	Stockout mitigation options	Operations manager	High
UC3	Approve and execute mitigation	Operations manager	★ riskiest
UC4	Partial-failure recovery	Platform on-call	High

3. Risks and guards

Risk	Guard
Duplicate or unauthorised action	Agents have no write tools; `GR-04` approval policy; `GR-05` one effect per idempotency key
Acting on stale stock	`GR-02` blocks snapshots > 15 min and re-reads before commit; version check on write
Runaway agents or cost	`GR-06` per-agent tool scopes, call, token and time budgets; dedup before any agent runs

Key assumption: inventory data is fresh (≤ 15 min) in every pilot warehouse. The Head of Inventory owns that SLA.

4. Evals

Measure	Metric	Baseline → target / threshold
Value	Event → plan · SLA misses on affected priority orders	26 → ≤ 2 h · 18 → ≤ 8 %
Quality	Impacted IDs exact · incidents resolved · safety invariants	100 % · ≥ 80 % · 100 %
Adoption	Plans approved without rework	≥ 70 %

Evidence so far: on 20 gold incidents, impact was exact 20/20, there were 0 duplicates under replay and fault injection, and 16/20 were resolved. Business impact is still a hypothesis. **Decision asked:** the **COO** approves a **10-week pilot** (shadow mode, then live reservations in one warehouse, ≈ €60k) by **2026-12-18-17-00-00 UTC**.